Lesson 8: Capstone: Battle Arena

Module 2 | 60 minutes | game design, functions, data | Grades 7–11 · ages 12–16

Name: _____ Class: _____ Date: _____

Plan your project and practise the key ideas.

1. Key words

Match each word to what it means. Write the letter.

a. HP ___ A. Give the bigger or smaller of two numbers; used to keep values in limits.
b. turn-based ___ B. The loop that repeats each turn until the game ends.
c. game loop ___ C. Players take turns, one action at a time.
d. max() / min() ___ D. Hit points: a fighter's health. At 0, they are defeated.

2. Plan your program

Before you type it: Battle helpers. Write attack(attacker, defender, damage) that takes damage off defender["hp"] (never below 0) and returns the defender's new HP. Write heal(fighter, amount) that adds amount to fighter["hp"] (never above fighter["max_hp"]) and returns the new HP. Write the steps in plain English (pseudocode), then the Python you will use.

3. Predict the output: the last hit

What does this program print? Code: hp = 20 ⏎ for dmg in [5, 8, 9]: ⏎ hp = max(0, hp - dmg) ⏎ print(hp)

a. A. 0
b. B. -2
c. C. 2
d. D. 20

4. Plan a battle turn

Put one turn of the battle in order Number them 1–6.

a. ___ Add 1 to the turn counter
b. ___ If the enemy still has HP, it hits back
c. ___ Do the player's move (attack or heal)
d. ___ Ask the player for a move
e. ___ Show both fighters' stats
f. ___ Loop again while both have HP left

5. Plan your program

Before you type it: Debug it! This mini battle should keep hitting the goblin until its HP is 0, then print `The goblin is defeated in ... turns!`. Right now it never ends. Fix the two bugs, and make sure enemy_hp never goes below 0. Write the steps in plain English (pseudocode), then the Python you will use.

6. My plan

Plan your project: Design and build a turn-based text battle game where your hero fights a random enemy using attack and heal, with stats in dictionaries and a function for every action. Sketch it and list what each part will do.

7. Checklist

Tick each one when your project does it.

a. ☐ Player and enemy stats in dictionaries, with a list of at least 3 enemies
b. ☐ Functions for damage and healing that return numbers within limits
c. ☐ Random damage and a randomly chosen enemy
d. ☐ A game loop that runs while both fighters have HP
e. ☐ attack and heal moves, limited potions and an Unknown move message
f. ☐ Victory and defeat endings, then GAME OVER

8. Two stars and a wish

Ask a friend to test your project. Write two things they liked and one idea to improve it.

How did it go? Colour a face: 😀 I can do it · 🙂 Nearly there · 🤔 I need help

☐ I can store game stats in dictionaries and a list.
☐ I can write functions that return damage and healing within limits.
☐ I can build a game loop with win and lose endings.
☐ I can playtest my game and balance its numbers.